

Walk Forward Trend Following and BTC Dominance

Mean Reversion in Cryptocurrency Markets

Peter Prendergast | Student ID: 25217979
University College London
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1 Data Download and Visualisation

I obtained OHLCV data for four cryptocurrency asset pairs, specifically BTC/USDT, ETH/USDT, BNB/USDT, and ADA/USDT. The dataset is retrieved from Binance using the `ccxt` Python library. To align with the available FRED Federal Funds Rate series, the sample period runs from 1 January 2020 to 20 March 2026, resulting in a final sample of 2,271 daily observations. Daily frequency represented a compromise between capturing medium term momentum dynamics and maintaining manageable data volumes.

Strategy 2 draws on a broader universe of 11 Binance spot pairs (BTC, ETH, BNB, SOL, XRP, ADA, AVAX, DOT, LINK, LTC, MATIC) to construct a diversified altcoin basket and a BTC dominance signal.

1.1 Data Cleaning

Each series is passed through a data cleaning pipeline that checks for duplicate timestamps, negative prices or trading volume, and inconsistencies in OHLCV bars. No structural issues are detected in any of the asset series, so the number of observations remains unchanged. Outliers are flagged for subsequent diagnostic analysis but are retained at this stage.

1.2 Excess Returns

The risk-free rate r_t^f is based on the FRED Effective Federal Funds Rate (series DFF), converted from an annual percentage rate into a daily compounded rate:

$$r_{\text{daily}}^f = \left(1 + \frac{\text{DFF}}{100}\right)^{1/365} - 1. \quad (1)$$

After forward-filling the rate across calendar days to align with the asset return index, excess returns are computed as $r_t^e = r_t - r_{t-1}^f$.

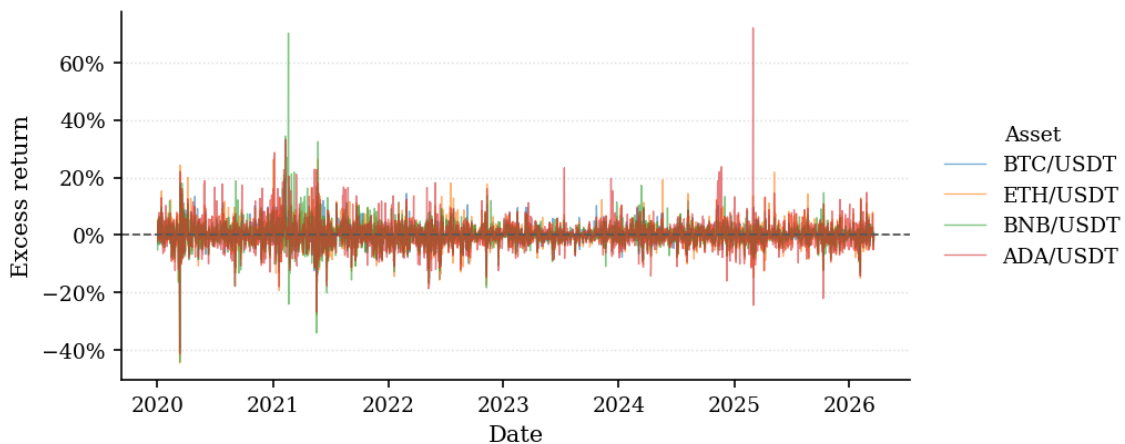


Figure 1: Daily excess return series for BTC/USDT, ETH/USDT, BNB/USDT, and ADA/USDT.

At daily frequency, the per-period risk-free rate is economically negligible relative to cryptocurrency price

movements. For example, a 5% annual rate is only about 0.014% per day, compared with typical daily crypto volatility of 3%–5% or more. As a result, subtracting r_t^f has almost no effect on returns, signals, or backtest results at this horizon.

2 Strategy

Both strategies are implemented using the same period for training and testing. The initial portfolio value is set at $V_0 = \$10,000$, subject to a firm gross exposure constraint of $\sum_i |\theta_{t,i}| \leq \$100,000$.

2.1 Strategy 1: Trend-Following

2.1.1 Rationale

As outlined in [Liu and Tsyvinski, 2021], cryptocurrency assets, similarly to other asset classes exhibit time series momentum behaviour. The intuition being that assets which have had positive excess returns tend to sustain that trend for some time in bull markets, the opposite is also observed with negative excess returns during bear markets.

2.1.2 Signal Construction

For each asset i at time t , I compute a volatility-normalised trend signal. This is done in two stages. First, I compute the raw trend $\tau_{i,t}^{\text{raw}}$, which captures the deviation of the close price $p_{i,t}$ from its w -bar moving average:

$$\tau_{i,t}^{\text{raw}} = \frac{p_{i,t}}{\text{MA}_{i,t}(w)} - 1. \quad (2)$$

where $\text{MA}_{i,t}(w) = \frac{1}{w} \sum_{k=0}^{w-1} p_{i,t-k}$. I standardise this signal by dividing it by the rolling standard deviation of simple returns over a w_{vol} -bar window:

$$z_{i,t} = \frac{\tau_{i,t}^{\text{raw}}}{\sigma_{i,t}(w_{\text{vol}})}. \quad (3)$$

The z-score is clipped and the position direction is determined by a threshold with a dead zone to prevent noisy oscillations from needlessly increasing costs.

2.1.3 Position Sizing using Mean Variance Optimisation

For position sizing I implement MVO using shrinkage estimation of the rolling covariance matrix [Ledoit and Wolf, 2004] with cvxpy quadratic programming.

On bars where the signal is active ($|z_{i,t}| > \delta$), the raw weight is $w_{i,t}^{\text{raw}} = \tau_{i,t}^{\text{raw}}$, whereas for inactive bars it is zero.

The optimisation step solves

$$\max_w \mu^\top w - \frac{\gamma}{2} w^\top \hat{\Sigma}_t w \quad \text{subject to} \quad \|w\|_1 \leq 1,$$

with $\gamma = 1$, where μ is proxied by the volatility scaled signal and $\hat{\Sigma}_t$ is the Ledoit Wolf shrinkage covariance estimate. While a diagonal inverse volatility approximation performed better in sample, the full MVO specification was preferred because it captures cross-asset covariance and should be more robust when correlations spike in stressed markets. Positions are rebalanced every n days, a configurable frequency.

2.1.4 Parameter Selection

The parameters for the strategy were selected using a walk forward validation procedure. An untouched test window of 126 consecutive daily observations (about 4.1 calendar months in a market that trades seven days per week) was kept for out of sample testing. This process used folds of 600 bars and a validation window of 100 bars, advancing by 100 bars per fold. This yielded 15 walk-forward folds of which every second were evaluated.

The search grid used the following parameters: moving-average window w , volatility window w_{vol} , dead-zone threshold δ , rebalancing frequency n , and gross utilisation u . The signal clip was fixed at ± 5 , and

the gross cap at \$100,000.

Each candidate combination is evaluated across all folds and the sharpe ratio is computed. The combinations are then selected based on a stability penalised mean validation Sharpe:

$$\text{score} = \bar{S}_{\text{val}} - \lambda \text{std}(S_{\text{val}}). \quad (4)$$

where $\bar{S}_{\text{val}}$ is the mean Sharpe ratio across folds, and $\text{std}(S_{\text{val}})$ is the cross-fold standard deviation. I set $\lambda = 0.1$ as the stability penalty. A candidate must have a positive mean Sharpe, and at least 55% of folds must exhibit positive Sharpe ratios.

This procedure selects $w = 140$ days, $w_{\text{vol}} = 30$ days, $\delta = 1.0$, $n = 10$ days, and $u = 0.10$. This combination yields a mean Sharpe of 0.464, with 62.5% positive folds as well as an adjusted score of 0.306. This long moving average window is consistent with the findings of [Moskowitz et al., 2012].

2.2 Strategy 2: BTC Dominance Mean-Reversion

2.2.1 Rationale

Bitcoin is the largest asset among cryptocurrencies, making up the majority of the market cap of the cryptocurrency market [CoinMarketCap, 2026]. The signal in this strategy sets BTC dominance as BTC's share of aggregate notional trading volume across cryptocurrencies. During regimes of fear and uncertainty this dominance tends to spike as crypto investors de risk into the most liquid crypto asset. The mean reversion strategy buys a basket of altcoins, anticipating a rotation back into these assets as the regime shifts.

2.2.2 Signal Construction

The signal is built from a notional dominance ratio $D_t = \frac{\text{notional}_t^{\text{BTC}}}{\sum_j \text{notional}_t^j}$. The notional is defined as close price times volume for each asset.

The first indicator for the strategy is a 20 day z-score of the dominance level:

$$z_t^{\text{dom}} = \frac{D_t - \bar{D}_t^{(20)}}{\hat{\sigma}_{D,t}^{(20)}}. \quad (5)$$

The second indicator is a 3 day rate of change of dominance $ROC_t = D_t - D_{t-3}$ which serves as a way to prevent the strategy getting caught in long-term market shifts.

Entry to the strategy happens when both of these metrics exceed their 80th and 95th 126 day rolling percentile respectively. The position exits when the dominance score drops to zero or a holding period of 7 days passes. I did experiment with a crash filter however the Optuna search consistently favoured the no filter configuration.

2.2.3 Parameter Selection

The signal parameters were tuned using Bayesian hyperparameter optimisation using Optuna's Tree structured Parzen Estimator (TPE) sampler with 80 trials. The search space covers: entry percentile, exit z-score, maximum holding period, ROC percentile, crash threshold and gross utilisation.

The walk forward scoring function is:

$$\text{score} = \bar{S}_{\text{val}} - 0.15 \cdot \text{std}(S_{\text{val}}) - P_{\text{inactivity}} \quad (6)$$

3 Transaction Costs

For transaction costs, I use the Abdi-Rinaldo high low spread estimator [Abdi and Rinaldo, 2017]. This estimator exploits the entire information content of OHLC bars. This provides an effective and stable measure for my daily frequency strategies.

The estimator is:

$$\hat{s}_t^2 = 4 (\ln c_{t-1} - m_{t-1}) (\ln c_t - m_t) \quad (7)$$

where c_t is the closing price and $m_t = (\ln h_t + \ln l_t)/2$ is the log midpoint. This is smoothed via a 21-day rolling mean to compute $\hat{s}_t = \sqrt{|\hat{s}_t^2|}$. These costs are applied at each bar. I chose this estimator as it remains well defined given all market conditions.

3.1 Cost Sensitivity

The cost-sensitivity analysis shows that Strategy 1 is much more resilient to trading costs than Strategy 2. Even when the estimated half spread is tripled, Strategy 1 still maintains a positive net Sharpe ratio and performance only weakening gradually. Strategy 2 is much more vulnerable. Its Sharpe falls quickly as costs rise and turns negative when the half spread is doubled. This fits the way the strategies are built since Strategy 1 trades less often and with lower turnover, while Strategy 2 uses a levered basket of altcoins where wider spreads and repeated rebalancing make slippage much more damaging.

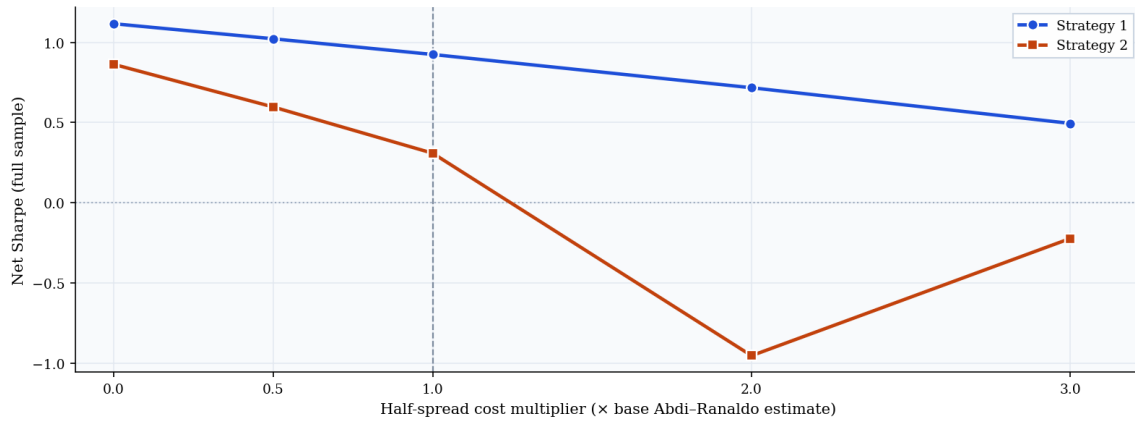


Figure 2: Transaction Cost Sensitivity Analysis

4 Performance

Strategy 1 keeps total exposure fixed at \$10,000 throughout the backtest, regardless of portfolio gains or losses. This maintains stable daily dollar risk, although effective leverage relative to portfolio value changes over time. This means that the maximum one-day profit or loss is bounded by fixed exposure multiplied by the largest absolute asset return on that day.

Strategy 2 is designed to trade at a target exposure of \$20,000. In normal conditions, that is the amount the strategy puts to work. The \$100,000 gross exposure limit is a set constraint, while the $10 \times V_t$ term acts as a safety check so the strategy never takes on more than 10 times its current equity. In practice, this means the strategy usually trades with a fixed \$20,000 exposure, but if the portfolio value falls far enough, the position size is automatically reduced to avoid excessive leverage.

4.1 Metrics

Both strategies deliver positive risk-adjusted returns over the development sample, although both underperform BTC buy-and-hold in total return. Their main advantage is improved downside control. Strategy 1 achieves a maximum drawdown of -22.3% , less than one-third of BTC buy-and-hold's -76.6% , while its Sortino ratio of 1.151 is essentially in line with the benchmark's 1.146. This suggests that the trend-following signal captures a meaningful share of upside while avoiding much of the benchmark's extreme downside exposure. Strategy 2 also produces positive in-sample performance, but its drawdown of -59.9% remains high for a mean-reversion strategy, largely because the selected specification uses $2 \times$ leverage. Trading costs are modest for Strategy 1 because of its low utilisation and infrequent rebalancing, whereas Strategy 2 is structurally more expensive due to its leveraged basket construction and the wider spreads of mid-cap altcoins.

The original holdout strengthens the case for Strategy 1. It records a Sharpe ratio of 1.30 with a shallow

Metric	Strategy 1		Strategy 2	
	IS	OOS	IS	OOS
Sharpe Ratio	0.923	1.300	0.343	-2.912
Sortino Ratio	1.151	1.838	0.197	-1.377
Calmar Ratio	0.825	3.324	0.105	-1.927
Ann. Return (%)	18.38	12.56	6.30	-14.0
Max Drawdown (%)	-22.3	-3.8	-59.9	-7.3
Total Net PnL (\$)	+31,903	+2,431	+6,818	-1,222
Gross PnL (\$)	+42,332	+2,511	+30,866	-596
Total Turnover (\$)	1,253,508	13,503	2,260,369	80,079
Return on V_0 (%)	319.0	24.3	68.2	-12.2
Active Days (%)	93.5	100.0	9.0	2.4
Avg. Holding Period (days)	10	10	3.6	1.5

Table 1: Original submitted performance metrics for both strategies (net of costs). Ratios use the project’s 252-observation annualisation convention. Strategy 1 holdout start equity: \$41,903. Strategy 2 is tuned with $u = 2.0$ (\$20,000 gross).

maximum drawdown of only -3.8% , while BTC buy-and-hold over the same window returns -26.2% with a Sharpe of -1.16 . The exposure breakdown shows that BTC accounted for the vast majority of gross exposure during this period. Strategy 2 instead lost money as the expected altcoin rebound did not materialise. Its two qualifying holdout trades are too few for a precise performance estimate, but the negative economic result is retained without retuning.

4.2 Continuous Post-Selection Evidence

Both frozen specifications were subsequently run through 4 August 2026. Covering the dates of the original 126-observation holdout and 137 completed post-freeze daily observations gives one uninterrupted 263-observation period from 15 November 2025. The metrics are recomputed from one stateful run of the hashed frozen evidence pipeline, not by adding rounded values from the submitted table. This longer window does not replace the original holdout: because its first segment was already observed when the strategies were frozen, it is explicitly treated as post-selection evidence.

Metric	Strategy 1	Strategy 2
Net Return (%)	+10.06	-19.44
Annualised Return (%)	+9.62	-18.71
Sharpe Ratio	1.192	-2.231
Maximum Drawdown (%)	-4.96	-19.44
Gross PnL (\$)	+4,563	-1,365
Transaction Costs (\$)	-336	-1,782
Net PnL (\$)	+4,227	-3,147
Total Turnover (\$)	90,823	282,099
Mean Gross Exposure (\$)	9,953	1,065
Activity	100.0% of days	7 entries / 14 days

Table 2: Continuous post-selection results from 15 November 2025 to 4 August 2026, net of estimated costs. Annualised figures use 252 observations per year for comparability with the submitted analysis.

Strategy 1 returned 10.06% with a Sharpe ratio of 1.192 and a -4.96% maximum drawdown, while BTC buy-and-hold and the equal-weight four-asset basket returned -32.23% and -42.29% , respectively. The result remains concentrated: mean absolute BTC exposure was \$7,718 of \$9,953 mean gross exposure. Strategy 2 returned -19.44% and lost \$1,365 before \$1,782 of estimated costs. Its mean exposure across all observations is low because it was active for only 14 days; when active, gross exposure remained approximately \$20,000.

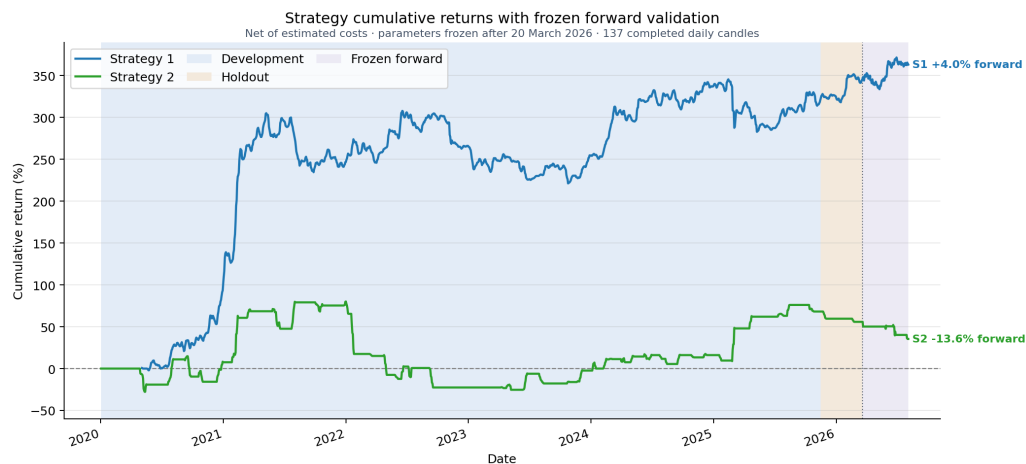


Figure 3: Submitted cumulative-return paths extended through 137 post-freeze observations. Shading separates development, original holdout, and frozen-forward evidence.

5 Next Steps

Both strategies remain regime dependent. Strategy 1's positive result is heavily concentrated in BTC, so a pre-specified expansion of the asset universe could improve diversification. The 263-observation post-selection period improves on the original short holdout but covers only about 8.6 calendar months, while Strategy 2 generated just seven entries and remained clearly negative. Further research should therefore extend frozen monitoring rather than retune the failed specification. Any regime filter, more conservative leverage rule, or combined capital-allocation framework should be fixed before additional observations arrive and evaluated as a separate experiment.

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